SafeBite: An AI-Driven Mobile Application for Personalized Food Ingredient Analysis and Health and Nutrition Recommendations Using RAG, LLMs, and Contextual Intelligence

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Software Requirements Specification

BY

Amjad Ali Abro | CMS ID: 023-23-0043

Sarfraz Ali Katpar | CMS ID: 023-23-0045

Sehrish Khemani | CMS ID: 023-23-0224



Supervised by: Dr. Saif Hassan

Co-Supervised by: Prof. Dr Sher Muhammad Daudpota

DEPARTMENT OF COMPUTER SCIENCE

FACULTY OF SCIENCE AND INFORMATION TECHNOLOGY

SUKKUR IBA UNIVERSITY

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1. Introduction

1.1. Purpose of Document

This Software Requirements Specification (SRS) defines the expected behavior, boundaries, interfaces, and quality requirements of SafeBite, a mobile application for personalized food ingredient analysis and health and nutrition recommendations. It provides a shared reference for the project team, supervisor, co-supervisor, evaluators, and future testers.

The document describes what the system is required to do rather than prescribing every implementation detail. Requirements are written to support design, development, verification, and review. The SRS reflects the current proposed scope and should be updated if the project scope is formally changed.

1.2. Intended Audience

The primary audience includes the FYP supervisor and co-supervisor, the three student developers, departmental evaluators, and testers. The document may also be used by future maintainers or users who need to understand the system's capabilities and limitations.

Readers reviewing the implementation should pay particular attention to the functional requirements, use-case scenarios, safety safeguards, privacy controls, and assumptions about external product and nutrition data.

1.3. Document Convention, Abbreviations, and Acronyms

Unless otherwise specified, requirements use the word “shall” for a mandatory requirement, “should” for a recommended requirement, and “may” for an optional capability. Requirement identifiers (for example, FR-01 and NFR-01) are intended to remain stable during review.

Document convention: Times New Roman is used throughout, with numbered section headings, bold labels for structured information, and tables for use cases and requirement summaries. The department’s section order and headings are retained.

Table 1: Abbreviations and Acronyms

Abbreviation	Full Form
AI	*Artificial Intelligence*
API	*Application Programming Interface*
OCR	*Optical Character Recognition*
LLM	*Large Language Model*
RAG	*Retrieval-Augmented Generation*
PHSS	*Personalized Health Suitability Score*
UI	*User Interface*
HTTPS	*Hypertext Transfer Protocol Secure*
FR	*Functional Requirement*
NFR	*Non-Functional Requirement*

2. Overall System Description

2.1. Project Background

Food packages contain ingredient lists, nutrition panels, allergen statements, and product identifiers, but the information can be difficult to interpret quickly. Users with food allergies, dietary preferences, or health-related nutrition concerns may need to compare several products and understand unfamiliar ingredient names. Product information may also be spread across package labels and online databases, and these sources do not always match.

SafeBite is proposed as a mobile application that brings product lookup, label text extraction, ingredient information, personalized avoid-list checks, and nutrition-related explanations into one workflow. It also includes an optional report-scanning feature that extracts lipid-profile values for the user to review. The application is intended to support informed choices, not to diagnose medical conditions or replace professional advice.

2.2. Project Scope

SafeBite will provide a mobile interface through which a user can sign in, manage a food profile, maintain a personal avoid list, and analyse packaged food products. Product analysis may begin with a barcode scan, a package-label image, or both. Where available, the application will retrieve a product record and use OCR to extract visible label text. The system will normalize and compare the available fields, flag mismatches or uncertain extraction, and check ingredients against its knowledge base, allergen rules, and the user's avoid list.

The application will retrieve relevant nutrition information and evidence, calculate a Personalized Health Suitability Score (PHSS) using documented rules, and present a result with warnings, source references, and uncertainty indicators. Users will be able to explore ingredient explanations, compare products in the context of their profile, and review or delete saved scan history.

An optional, separate health-report workflow will allow the user to consent to report processing, capture a report, review extracted lipid-profile values and units, correct errors where supported, and choose whether to save the confirmed values. The system will not assume that extracted report values are medically verified.

Core modules include:

- Account and privacy management
- Food profile and personal avoid list
- Hybrid barcode lookup and package-label OCR
- Ingredient normalization, knowledge lookup, and allergen checks
- Nutrition evidence retrieval, PHSS calculation, and result presentation
- Ingredient Explorer and personalized product comparison
- Optional health-report and lipid-profile extraction
- Scan history, review, and deletion

2.3. Not in Scope

The following items are outside the current project scope:

- Medical diagnosis, treatment planning, or medication recommendations.
- Replacing a doctor, dietitian, laboratory, or certified food-safety authority.

- Guaranteeing that a product is safe solely from its barcode, OCR output, or an external database record.
- Automatic connection to hospital systems, laboratory information systems, or electronic health records unless separately approved and technically available.
- Continuous monitoring of a user's health, wearable devices, or real-time physiological measurements.
- A guarantee of complete coverage for every product, brand, country, language, or ingredient database.
- A fully autonomous AI agent; SafeBite is a user-directed application that uses AI components for defined analysis and explanation tasks.

2.4. Project Objectives

- Provide a mobile workflow for reviewing packaged food ingredients and available nutrition information.
- Combine barcode-based product retrieval with OCR-based label extraction to improve the available product information.
- Identify ingredients that match a user's avoid list or known allergen preferences and clearly display relevant warnings.
- Provide ingredient explanations grounded in retrieved reference information, using RAG and an LLM where appropriate.
- Calculate and display PHSS according to documented, testable rules, while making its limitations clear.
- Support side-by-side comparison of products based on available ingredients, nutrition fields, and the user's configured preferences.
- Extract lipid-profile values from user-provided reports as an optional feature, requiring user review before the values are used or saved.
- Protect personal profile and report information through consent, access controls, and user-managed history.
- Evaluate the system for functional correctness, extraction quality, evidence grounding, usability, and safe handling of uncertain information.

2.5. Stakeholders

a) **End Users:** People who want help understanding packaged-food ingredients, allergen statements, or nutrition information.

b) **Users with Dietary Restrictions:** People who configure avoid lists or allergen preferences; the system must not imply that these preferences are a medical diagnosis.

c) **FYP Team:** Amjad Ali Abro, Sarfraz Ali Katpar, and Sehrish Khemani, responsible for requirements, design, development, testing, and documentation.

d) **Supervisor:** Dr. Saif Hassan, who provides academic supervision and reviews project progress and deliverables.

e) **Co-supervisor:** Prof. Dr. Sher Muhammad Daudpota, who provides additional technical and academic guidance.

f) **Departmental Evaluators:** Assess the project against academic objectives and the approved scope.

g) **External Service and Data Providers:** Product data sources, hosting providers, authentication services, OCR/LLM services, and nutrition references, depending on the final implementation.

h) **Future Maintainers:** Developers who may update the knowledge base, dependencies, or application after the FYP.

2.6. Operating Environment

Client Platform: A mobile application intended primarily for Android smartphones. Support for iOS is dependent on available development time and the final deployment approach.

Device Capabilities: A working rear-facing camera is needed for barcode and label capture; report scanning also uses the camera. A device with sufficient memory and a supported operating system version is required.

Server Environment: A backend API will process requests, coordinate product lookup and OCR/LLM-related services, apply rule-based checks, and access the application's database and knowledge resources. The exact deployment provider will be selected during implementation.

Network: An internet connection is expected for account operations, external product lookup, backend processing, and retrieval of hosted knowledge resources. If a service is unavailable, the app should communicate the limitation rather than fabricate a result.

Proposed technology choices include Flutter for the mobile interface, a Python-based API such as FastAPI, OCR tooling such as PaddleOCR or EasyOCR, a barcode decoding library, a relational database such as PostgreSQL, and an embedding/retrieval component such as Sentence Transformers with FAISS. These are implementation proposals and may change after feasibility testing.

2.7. System Constraints

Software Constraints: OCR and barcode decoding depend on image quality, label layout, supported formats, and library capabilities. External API limits and service availability may affect product retrieval and response times.

Hardware Constraints: Camera focus, lighting, glare, small print, damaged packaging, and device performance can reduce extraction quality.

Language Constraints: Initial implementation may prioritize English-language ingredient lists and nutrition labels. Support for other languages will depend on OCR and knowledge-base coverage.

Data Constraints: Product databases may be incomplete, outdated, region-specific, or inconsistent with the physical package. SafeBite must flag conflicts and should prefer visible package evidence when the user can review it.

Safety Constraints: A PHSS is a decision-support indicator, not a clinically validated medical score. A detected allergen conflict must not be hidden by a high numerical score.

Privacy Constraints: Health reports and profile data are sensitive. The application must request explicit consent before processing a report and must not save report values without a clear user's choice.

Third-party Constraints: External APIs, hosted models, OCR packages, and data sources may have usage limits, licensing conditions, or availability restrictions.

Academic Constraints: Features and evaluation must remain feasible within the FYP timeline and must be demonstrated with documented test cases.

2.8. Assumptions & Dependencies

- Users have a compatible smartphone and can grant camera permission when they choose to scan a product or report.
- Users provide accurate profile and avoid-list information, and review extracted information before relying on it.
- A product-data source such as Open Food Facts, and/or another approved source, is accessible for at least a portion of test products.
- A nutrition reference such as USDA FoodData Central may be used where suitable; the final sources and licensing will be documented.
- OCR, barcode, embedding, and LLM services or libraries remain available under acceptable terms during development.
- The backend and database can be deployed in a supported environment with appropriate secrets management and network access.
- The PHSS formula, feature weights, and interpretation bands will be documented and approved by the project team and supervisor before evaluation; they are provisional until validated.
- The application will present an uncertainty or unable-to-verify state when required product data, extraction confidence, or evidence is insufficient.

3. External Interface Requirements

SafeBite interacts with the smartphone camera, mobile operating system, backend API, product and nutrition data sources, database, and optional OCR/LLM or retrieval services. The high-level interface context is summarized below.

Table 2: External Interface Requirements

Interface component	Information exchanged
User / Smartphone	Camera images, barcode, profile input, consent and confirmations
SafeBite Mobile App	User interface, capture, result display, profile and history controls
SafeBite Backend API	Validation, workflow coordination, PHSS rules, access control
External Product / Nutrition Sources	Product fields and available nutrition information
OCR / Retrieval / LLM Components	Extracted text, retrieved evidence, grounded explanations
Database	Profiles, preferences, saved product scans and user-approved report values

3.1. Hardware Interfaces

Smartphone Camera: Captures barcode images, package ingredient/nutrition labels, and optional health-report pages. The application shall request camera permission at the point of use and provide a clear response if permission is denied.

Display and Touch Input: Present scan instructions, extracted fields, warnings, result details, consent choices, and confirmation controls.

Network Hardware: Wi-Fi or mobile data may be used for backend and external service requests.

Local Storage: Temporary image data may be held during processing. Persistent storage of report images or extracted values must follow the user's explicit choices and the implemented retention policy.

3.2. Software Interfaces

Mobile Operating System: Android camera, permission, secure storage, and network APIs. Exact minimum OS versions will be set during implementation.

Backend API: The mobile client sends authenticated requests for profile operations, product analysis, report extraction, comparison, and history management. Responses should use structured JSON with status and uncertainty fields.

Product Data Source: A barcode/product API, such as Open Food Facts if approved, may provide product name, brand, ingredients, nutrition fields, and product identifiers. Missing fields must remain marked as unavailable.

OCR Component: An OCR library or service extracts text from captured images and may return text regions or confidence values. The system shall not treat raw OCR text as verified data.

Barcode decoder: A compatible library such as ZXing or pyzbar may decode supported barcode formats from camera images.

Knowledge retrieval: An embedding model and vector index, such as Sentence Transformers with FAISS, may retrieve ingredient explanations and nutrition-related evidence.

LLM Service: A selected LLM may generate user-friendly explanations from retrieved evidence. The application must not rely on an unsupported model statement as the sole basis for an allergen decision.

Database: A relational database such as PostgreSQL may store accounts, preferences, product-scan metadata, and user-approved history. Sensitive report data should be access-controlled and stored only when the user chooses to save it.

Authentication Service: A selected authentication implementation shall provide account verification and session handling. The final provider and version will be documented before release.

3.3. Communications Interfaces

- The mobile application shall communicate with the backend through HTTPS for production or externally accessible deployments.
- Requests and responses should use UTF-8 JSON for structured application data; image uploads may use multipart/form-data or a documented equivalent.
- Authentication credentials and API secrets shall not be embedded as plain text in the mobile client or written to application logs.
- The client shall handle timeouts, unavailable services, invalid responses, and network disconnection with understandable messages and a retry option where appropriate.
- The system shall avoid sending report images or personal health context to external services unless required for the chosen workflow, covered by the privacy notice, and explicitly consented to by the user.
- Data transfer, authentication, and error responses shall be documented in the API specification during implementation.

4. Functional Requirements

4.1. Functional Hierarchy

SafeBite is organized into the following functional modules and sub-functions:

1. **Account and Profile Management:** Sign-in; view privacy information; manage food preferences; create and update the personal avoid list.
2. **Product Capture and Identification:** Scan a barcode; capture a package label; retrieve a product record; run OCR; normalize and compare extracted fields; flag low-confidence or conflicting data.
3. **Ingredient and Allergen Analysis:** Match ingredients to the knowledge base; check avoid-list and allergen rules; identify unknown ingredients; retrieve relevant evidence.
4. **Nutrition Assessment and PHSS:** Obtain available nutrition fields; apply documented scoring rules; apply allergen conflict overrides; show score, category, evidence, and uncertainty.
5. **Ingredient Explorer:** Show a plain-language explanation of a selected ingredient, supported by retrieved references and limitations.
6. **Personalized Product Comparison:** Compare two or more available product records against the user's configured preferences and available nutrition information.
7. **Health-Report Scanning:** Request separate consent; capture report image; extract lipid values and units; allow review and correction; save only if the user opts in.
8. **Scan History and Data Control:** List saved product scans and saved report values; open details; delete selected history records.
9. **Administration and Knowledge Maintenance:** Maintain supported ingredient references, scoring rules, and data-source configuration as permitted to the project team.

Table 3: Features and Requirements summary of the system

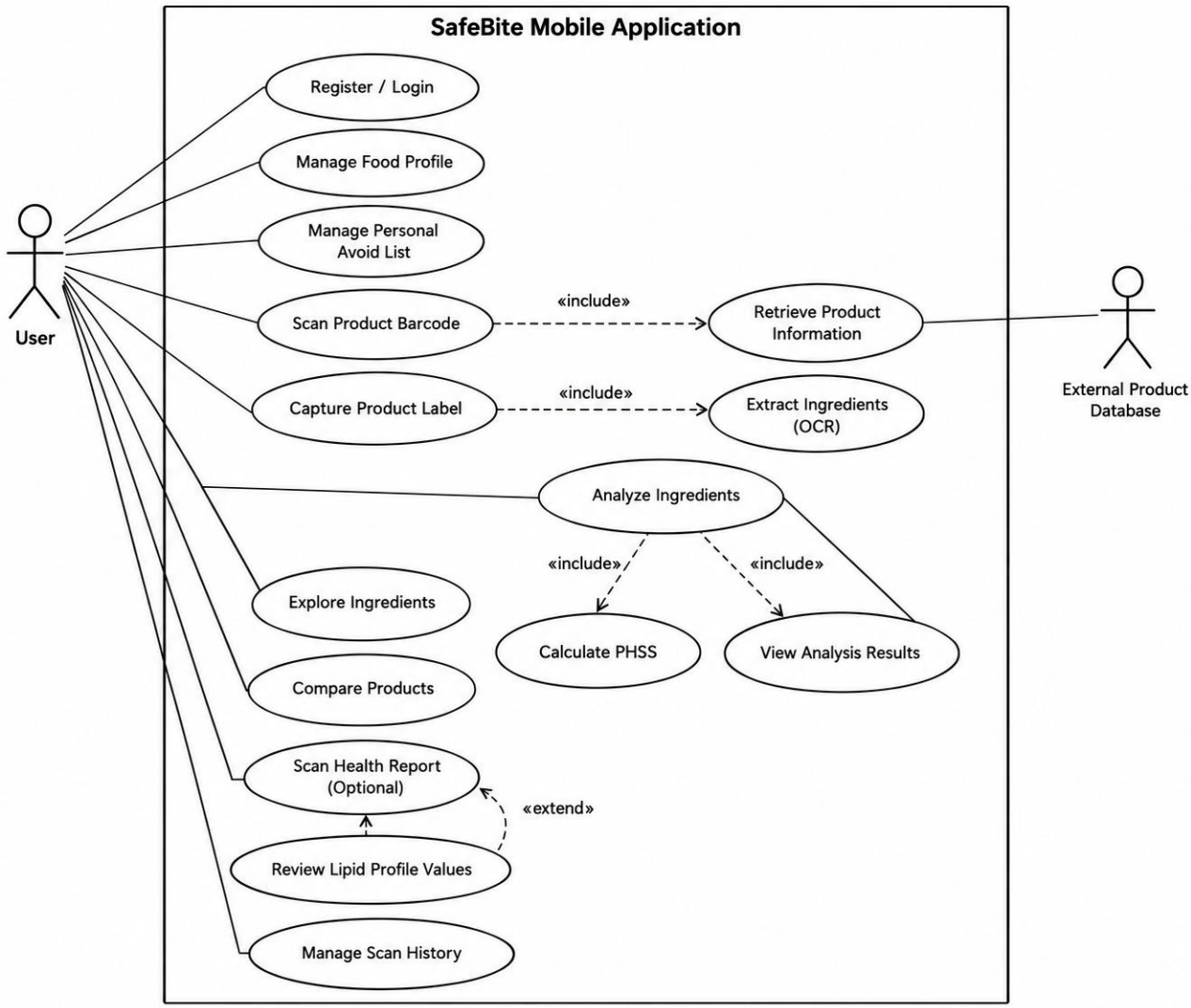
ID	Feature	Requirement summary
FR-01	Account and profile	The system shall allow a user to sign in and manage food preferences.
FR-02	Barcode and OCR	The system shall accept barcode input, package-label images, or both.
FR-03	Data reconciliation	The system shall normalize available fields and flag conflicts or low confidence.
FR-04	Ingredient and allergen checks	The system shall compare ingredients with the knowledge base and user avoid list.
FR-05	PHSS and results	The system shall calculate PHSS using documented rules and show warnings and sources.
FR-06	Ingredient Explorer	The system shall provide evidence-supported ingredient explanations where data is available.
FR-07	Product comparison	The system shall compare available product fields against the user's preferences.
FR-08	Report scanning	The system shall extract lipid values for user review after separate consent.

ID	Feature	Requirement summary
FR-09	History	The system shall allow users to review and delete saved records.
FR-10	Privacy choices	The system shall not save extracted report values unless the user chooses to save them.

4.2. Use Cases

The main external actor is the registered user. External systems—including product-data sources, OCR and retrieval components, the LLM service, and the database—support the use cases but do not replace the user's decisions. The following use cases describe the expected user-facing behaviour. The use-case diagram is depicted below:

Figure 1: Use Case Diagram of the SafeBite Mobile Application



4.2.1. Create or Update Food Profile

Create or Update Food Profile describes how the user completes this task and how SafeBite responds.

UC-01: Create or Update Food Profile		
Use case Id:	UC-01	
Actors: Registered user		
Feature: Account and profile management		
Pre-condition: The user has opened SafeBite and is signed in.		
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Open profile settings.	The system displays saved profile fields and privacy information.
2.	Enter or update dietary preferences and avoid-list items.	The system validates the entered fields.
3.	Review the changes.	The system shows a summary for confirmation.
4.	Save the profile	The system stores the changes for the authenticated user and confirms completion.
Alternate Scenarios:		
1a: If required fields are invalid, the system identifies them and does not save the invalid changes. **1b:** If the user cancels, the existing profile remains unchanged.		
Post Conditions		
Step#	**Description**	
1.	The profile is created or updated for the current user.	
2.	The saved preferences are available to product-analysis checks.	
Use Case Cross referenced	UC-02, UC-00	

4.2.2. Analyze a Packaged Food Product

Analyse a Packaged Food Product describes how the user completes this task and how SafeBite responds.

UC-02: Analyze a Packaged Food Product		
Use case Id:	UC-02	
Actors: Registered user; product-data source; OCR component; knowledge retrieval service		
Feature: Product capture, ingredient analysis, and PHSS		
Pre-condition: The user is signed in; camera permission is available if capturing images; network access is available for remote lookup.		
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Choose product analysis and scan a barcode, capture a label, or use both.	The app captures the selected input and validates that an image or barcode was provided.
2.	Submit the product input for analysis.	The system retrieves a matching product record where available and runs OCR on the package image.
3.	Review normalized product fields and any mismatch flags.	The system compares available sources and marks uncertain or conflicting fields.
4.	View the analysis result.	The system checks allergens and avoid-list rules, retrieves relevant evidence, calculates PHSS, and displays warnings, sources, and uncertainty.
Alternate Scenarios:		
1a: If no barcode record is found, the system continues with readable label data where possible. If extraction or evidence is insufficient, it displays an unable-to-verify state instead of claiming the product is safe.		
Post Conditions		
Step#	**Description**	
1.	The user receives a product analysis with visible limitations and warnings.	
2.	The user may choose to save the scan to history.	
Use Case Cross referenced	UC-01, UC-03, UC-04, UC-07	

4.4.3. Explore an Ingredient

Explore an Ingredient describes how the user completes this task and how SafeBite responds.

UC-03: Explore an Ingredient		
Use case Id:	UC-03	
Actors: Registered user; knowledge retrieval service; LLM service (if enabled)		
Feature: Ingredient Explorer		
Pre-condition: The user has opened an ingredient from a product result or search interface.		
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Select an ingredient.	The system identifies the selected ingredient and searches the approved knowledge base.
2.	Request an explanation.	The system retrieves relevant ingredient evidence and available references.
3.	Read the explanation and sources.	The app presents a plain-language summary, references, and any uncertainty or limitations.
4.	Return to the product result.	The system returns the user to the previous analysis without losing the result.
Alternate Scenarios:		
1a: If no reliable evidence is retrieved, the system states that sufficient information is unavailable and avoids inventing an explanation.		
Post Conditions		
Step#	**Description**	
1.	The ingredient information is shown with its available supporting sources.	
2.	The original product analysis remains accessible during the session.	
Use Case Cross referenced	UC-02	

4.2.4. Compare Products

Compare Products describes how the user completes this task and how SafeBite responds.

UC-04: Compare Products		
Use case Id:	UC-04	
Actors: Registered user; product-data source; knowledge retrieval service		
Feature: Personalized product comparison		
Pre-condition: The user is signed in and at least two products have usable records or scan results.		
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Select two or more products to compare.	The system loads the available ingredient and nutrition fields for each product.
2.	Start comparison.	The system aligns comparable fields and applies the user's saved preferences and avoid-list rules.
3.	Review the comparison.	The system shows differences, warnings, missing fields, and relevant references.
4.	Open a product detail if needed.	The system displays the selected product's analysis and limitations.
Alternate Scenarios:		
1a: If products have missing or incomparable fields, the system marks those fields as unavailable rather than assuming equal values. A known allergen warning remains prominent.		
Post Conditions		
Step#	**Description**	
1.	The user sees a side-by-side comparison based on available data.	
2.	No product is described as universally safe based only on a comparison score.	
Use Case Cross referenced	UC-01, UC-02	

4.2.5. Scan and Review a Health Report

Scan and Review a Health Report describes how the user completes this task and how SafeBite responds.

UC-05: Scan and Review a Health Report		
Use case Id:	UC-05	
Actors: Registered user; OCR component		
Feature: Optional health-report and lipid-profile extraction		
Pre-condition: The user is signed in and has reviewed the separate health-report privacy notice.		
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Choose report scanning and provide explicit consent.	The system records the consent decision for the current workflow and enables capture only after consent.
2.	Capture or select a report image.	The system extracts visible lipid-profile values and units where recognizable.
3.	Review and correct extracted values.	The system displays each extracted value with its unit and allows correction or discard.
4.	Choose whether to save confirmed values.	The system saves only if the user explicitly chooses to do so, then confirms the outcome.
Alternate Scenarios:		
1a: If consent is declined, processing does not begin. If a value or unit is unclear, it is marked for review. If the user declines saving, the values are not persisted as history.		
Post Conditions		
Step#	**Description**	
1.	The user has reviewed the extracted values or discarded them.	
2.	Only values explicitly approved for saving are stored in the user's history.	
Use Case Cross referenced	UC-01, UC-07	

4.2.6. Manage Personal Avoid List

Manage Personal Avoid List describes how the user completes this task and how SafeBite responds.

UC-06: Manage Personal Avoid List		
Use case Id:	UC-06	
Actors: Registered user		
Feature: Food profile and allergen/avoid-list management		
Pre-condition: The user is signed in.		
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Open avoid-list settings.	The system displays the current list and available edit controls.
2.	Add, edit, or remove an item.	The system validates the item and updates the pending list.
3.	Confirm the change.	The system saves the updated list and makes it available to product analysis.
4.	Open a product result later.	The system checks the result against the current saved list.
Alternate Scenarios:		
1a: If an entry is ambiguous, the system asks the user to clarify or stores it as a user-defined term without claiming a verified allergen match.		
Post Conditions		
Step#	**Description**	
1.	The avoid list reflects the user's confirmed changes.	
2.	Subsequent product checks use the updated list.	
Use Case Cross referenced	UC-01, UC-02	

4.2.7. Review or Delete Scan History

Review or Delete Scan History describes how the user completes this task and how SafeBite responds.

UC-07: Review or Delete Scan History		
Use case Id:	UC-07	
Actors: Registered user		
Feature: Scan history and data control		
Pre-condition: The user is signed in and has saved one or more product scans or report values.		
Scenarios		
Step#	**Action**	**Software Reaction**
1.	Open scan history.	The system lists the user's saved records and their dates or record types.
2.	Select a record.	The system displays the stored details permitted by the user's account.
3.	Choose delete and confirm.	The system deletes the selected record from the active history store.
4.	Return to history.	The system refreshes the list and confirms the record is no longer shown.
Alternate Scenarios:		
1a: If no records exist, the system displays an empty state. If deletion fails, the system explains that the action was not completed and does not falsely confirm success.		
Post Conditions		
Step#	**Description**	
1.	The user can review saved records.	
2.	A successfully deleted record is removed from the user's active history view and storage according to the implemented retention policy.	
Use Case Cross referenced	UC-02, UC-05	

5. Non-functional Requirements

5.1. Performance Requirements

- **NFR-PER-01:** For a normal network connection and a supported test device, the application should display a basic product-analysis result within 10 seconds for at least 80% of test cases, excluding unusually slow third-party service responses. This target will be measured during evaluation.
- **NFR-PER-02:** The application shall display progress feedback for operations that take noticeable time, including OCR, report extraction, retrieval, and product comparison.
- **NFR-PER-03:** The application shall not block the interface indefinitely when a backend or external service times out; it shall provide a failure message and a retry path.
- **NFR-PER-04:** PHSS calculation and avoid-list matching shall be deterministic for the same normalized input and the same configured rules.
- **NFR-PER-05:** The system should support at least 10 concurrent test sessions in a controlled demonstration environment, subject to the selected hosting plan and external service limits.
- **NFR-PER-06:** OCR quality and response time shall be evaluated on a documented test set of package labels and report images, including examples with varying lighting and image quality.

5.2. Safety Requirements

- **NFR-SAF-01:** The system shall not describe its output as a medical diagnosis or as a guarantee that a product is safe for every user.
- **NFR-SAF-02:** A known match to a user-configured allergen or avoid-list item shall be shown as a prominent warning and shall not be cancelled out by a high PHSS.
- **NFR-SAF-03:** If an ingredient list is missing, unreadable, contradictory, or insufficient to assess a critical restriction, the system shall display an uncertainty or unable-to-verify message.
- **NFR-SAF-04:** The system shall distinguish information extracted from a package, information retrieved from a database, and explanations generated from retrieved evidence wherever practical.
- **NFR-SAF-05:** Extracted lipid-profile values shall be shown with their units and shall require user review. The application shall not independently interpret them as a diagnosis or prescribe treatment.
- **NFR-SAF-06:** Health-related explanations shall encourage users to consult a qualified healthcare professional for clinical interpretation or urgent concerns.
- **NFR-SAF-07:** The system shall provide a way to correct or discard misread extracted values before they are saved.
- **NFR-SAF-08:** PHSS weights and category bands shall be documented and presented as project-defined decision-support rules unless clinically validated evidence supports a stronger claim.

5.3. Security Requirements

- **NFR-SEC-01:** The system shall require authentication before accessing private profiles and saved history.
- **NFR-SEC-02:** The backend shall enforce user-level authorization so that one user cannot access another user's profile, scans, or report values.

- **NFR-SEC-03:** Production communications shall use HTTPS, and passwords or authentication secrets shall not be stored in plain text.
- **NFR-SEC-04:** The application shall explain why a health report is being processed and obtain separate explicit consent before report scanning.
- **NFR-SEC-05:** The application shall ask whether confirmed lipid values should be saved and shall not persist them if the user declines.
- **NFR-SEC-06:** Users shall be able to review and delete saved history. The implementation shall define whether deletion is immediate and how backups or logs are handled.
- **NFR-SEC-07:** Logs shall avoid recording passwords, access tokens, full report contents, or unnecessary personal health information.
- **NFR-SEC-08:** API keys and service credentials shall be held in server-side configuration or a suitable secret store, not committed to source control.
- **NFR-SEC-09:** The application shall request only the device permissions required for a feature and explain camera access in context.
- **NFR-SEC-10:** The privacy notice shall describe the categories of data collected, purpose of processing, external service use where applicable, and available user controls.

5.4. User Documentation

- A concise user guide covering account setup, privacy choices, food-profile setup, and avoid-list management.
- A product-scanning guide explaining barcode scanning, label capture, OCR uncertainty, warnings, PHSS, and product comparison.
- A health-report guide explaining consent, report capture, lipid-value review, units, saving choices, and medical limitations.
- A troubleshooting guide for camera permission, poor image quality, missing product records, network errors, and unavailable services.
- A project technical document covering architecture, API interfaces, database structure, configuration, setup, and test procedures.

2. 6. References

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3. 7. Appendices

Appendix A — Provisional PHSS note: PHSS is a project-defined score intended to summarize the application's documented assessment rules. The exact formula, feature weights, category thresholds, and test methodology must be recorded in the implementation documentation and reviewed with the supervisor. A known allergen or avoid-list conflict takes precedence over the numerical score. The score must not be represented as a clinically validated measure.

Appendix B — Requirement verification: Each functional and non-functional requirement should be linked to at least one test case in the final test plan. Requirements should be reviewed with the supervisor and co-supervisor before implementation is treated as complete.